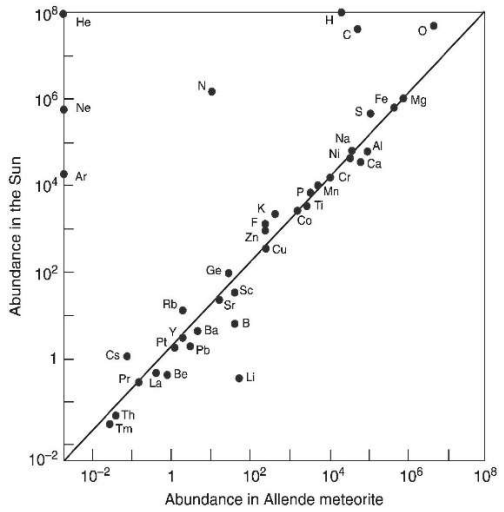
TOOLKIT: Chemistry of the Earth
Practical One: Chemistry of the whole Earth

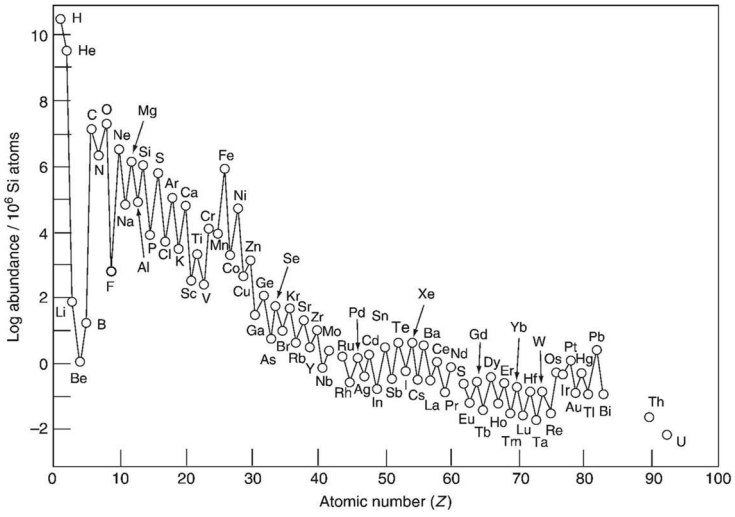
Objectives:

- Understand which elements are the most abundant in the Solar System, and why.
- Understand how the most abundant elements are distributed between the Earth's Core and Mantle.
- Calculate the chemistry of the whole Earth based on Core and Mantle compositions.
- Compare the composition of the whole Earth with the composition of the Solar System.

FEEDBACK

Abundances of elements in the Solar System





Introduction to Geochemistry: Principles and Applications, First Edition. Kula C. Misra.
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A. Which elements are the most abundant in the Solar System, and why?

1) Review the material from Lecture 1 in which we discussed nucleosynthesis and the origin of chemical elements. Table 1 lists the most abundant elements in the Solar System. Complete Table 1 to give the chemical symbols of these elements and their atomic numbers (Z).

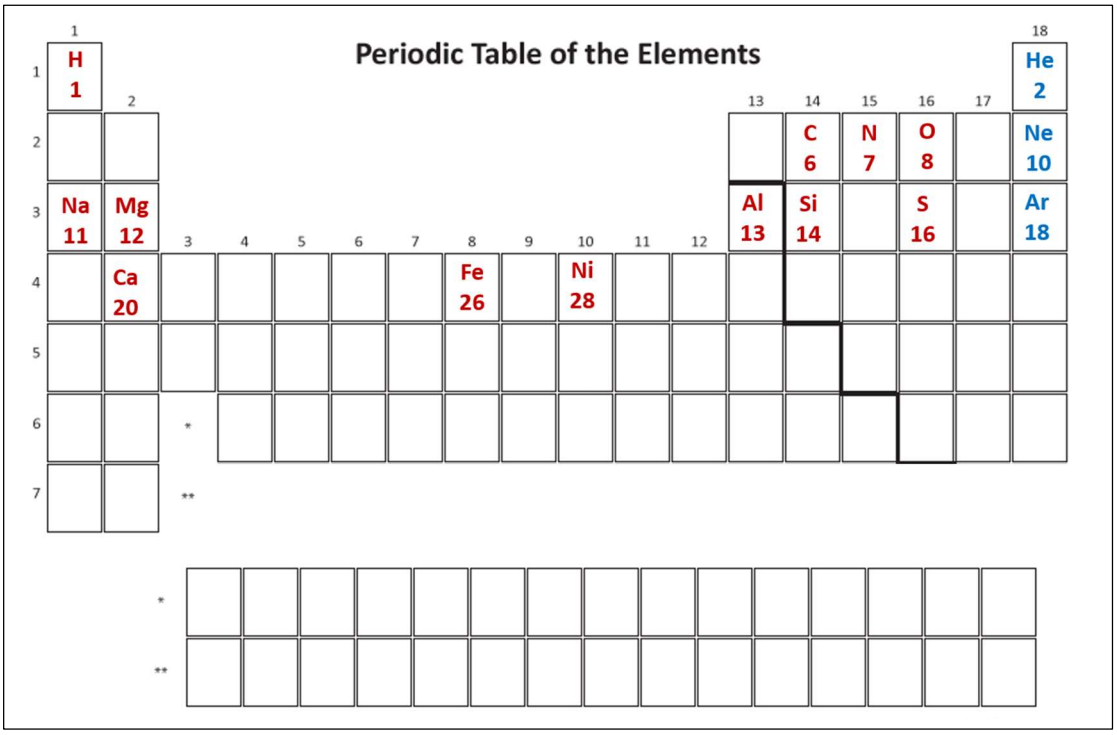
Table 1: Most abundant elements in the Solar System

Symbol	Name	Atomic number (Z)
H	Hydrogen	1
C	Carbon	6
N	Nitrogen	7
O	Oxygen	8
Na	Sodium	11
Mg	Magnesium	12
Al	Aluminium	13
Si	Silicon	14
S	Sulphur	16
Ca	Calcium	20
Fe	Iron	26
Ni	Nickel	28

2) The heaviest naturally occurring element is Uranium (U, Z = 92), whereas all the elements in the table have Z < 30. Why do the most abundant elements mostly have low atomic numbers?

This is the result of nucleosynthesis – lighter elements are produced in nuclear fusion reactions using H and He, the most abundant elements in stars.

3) In the Periodic Table below, locate the positions of the most abundant elements in the Solar System and fill in their chemical symbols and atomic numbers. The table is completed below. Remember to think about WHY these elements are the most abundant – nucleosynthesis favours low atomic numbers, even Z more common than odd Z, and there is special stability for Fe and Ni.



4) Table 1 does not include the Noble Gases helium, neon and argon, because today we are only concerned with elements that form solids. Add these elements to the Periodic Table above. Why are these elements abundant in the Solar System?

The noble gases are all in the final column of the Periodic Table. He, Ne and Ar are in high abundance in the Solar System - but not in the Earth. These elements are abundant because nucleosynthesis favours light elements, and even atomic numbers. Also, He formed in the Big Bang.

B. How are the most abundant elements distributed between the Earth's Core and Mantle?

Table 2: Major elements of the Mantle and Core		
Element	Mantle	Core
O	44.0	0
Na	0.27	0
Mg	22.8	0
Al	2.35	0
Si	21.0	6.0
S	0.03	1.9
Ca	2.53	0
Fe	6.26	85.5
Ni	0.20	5.2

The Table gives the weight percentages of elements in the Mantle and Core.

1) List the most abundant elements in the Mantle in order of abundance:

O, Mg, Si, Fe, Ca, Al, Na, Ni, S

2) Find an image of a **peridotite mantle nodule,** either online or in the lecture slides. Also, make sure to look at the examples provided. Either insert the image here, or draw a sketch of it. Then describe in words what material the Mantle consists of:

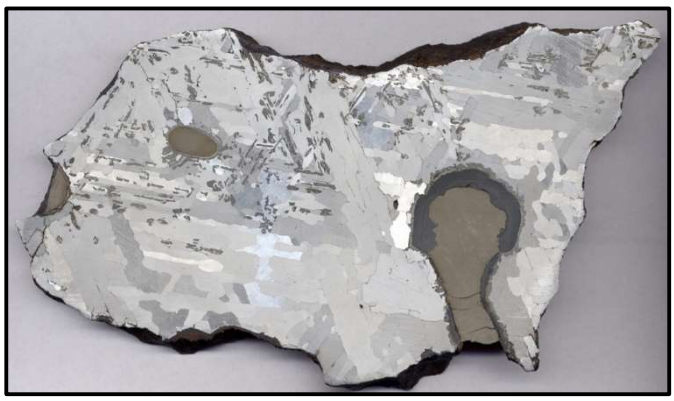


Mantle – peridotite, which mainly consists of green crystalline olivine and pyroxene – iron, magnesium silicate minerals. Note that O, Mg, Si and Fe are the most abundant elements in the mantle – no coincidence. Mantle nodules consist of peridotite nodules (rounded rocks) within dark grey basalt. The basalt ripped off chunks of the mantle as it ascended to the Earth's surface.

3) List the most abundant elements in the Core in order of abundance:

Fe, Si, Ni, S (note that in the lecture we talked about models in which the core contains oxygen)

4) Find an image of an **iron metorite,** either online or in the lecture slides. Also, make sure to look at the examples provided. Either insert the image here, or draw a sketch of it. Then describe in words what material the Core consists of:



Iron meteorites are pieces of iron-nickel metal. They are fragments of the cores of asteroids that underwent melting and core formation (this happened over 4.5 billion years ago). Iron meteorites have a criss-cross pattern of crystals that is only produced because they contain around 5% nickel. As the metal cools, it divides into two minerals, one Ni-rich and one Ni-poor, and the crystal structure is determined by this process.

Iron meteorites are our best analogue for the Earth's core, but the Earth's core is at much higher temperatures and pressures than are ever reached on a small asteroid. Also, the Earth's core has a solid inner layer and a liquid outer layer. So iron meteorites are similar, but not an exact match, to the core of the Earth.

C. Calculation of the chemistry of the whole Earth, based on Core and Mantle compositions.

1) You were given the composition of the Core and Mantle in Section B. In order to calculate the composition of the whole Earth, we also need to know the relative masses of the Core and Mantle.

Complete the following Table:

Table 3: Calculate the relative masses of the Core and Mantle						
	Radius km	Volume km^3	Volume m^3	Density kg/m^3	Mass kg	Mass percent
Earth	6371	1.08E+12	1.08E+21	5515	5.97E+24	
Core	3471	1.75E+11	1.75E+20	10915	1.91E+24	32
Mantle	XXX	9.08E+11	9.08E+20	4476	4.06E+24	68

Volume of a sphere: $V = \frac{4\pi r^3}{3}$ density = $\frac{\text{mass}}{\text{volume}}$

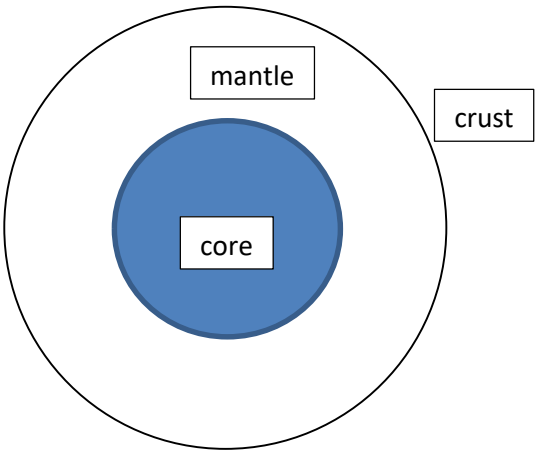
Show your working in the space below:

Comments:

Converting km^3 to m^3 means you need to multiply by 10^9 – not 10^3. A volume of 1 km^3 is a cube of dimensions 1000 x 1000 x 1000 m, i.e. 10^9 m^3.

You do not need to provide a radius for the mantle because it does not really have a radius. Calculate the volume of the mantle by subtracting the volume of the core from the volume of the Earth.

2) The average thickness of the Earth's Crust is 17 km. We will ignore the Crust in our calculation of the whole Earth composition because it is a relatively small volume. In the circle to the right, which represents a cross-section of the Earth, sketch the boundaries between Core, Mantle and Crust to show their relative sizes.



The core is approximately half the radius of the Earth. Check how accurately you have represented this. The width of the crust is so insignificant that it is about the width of the line.

3) Use the values of mass% Core and Mantle from above to calculate the composition of the whole Earth – complete column 4 in Table 4 below.

Table 4: Calculate the composition of the whole Earth and compare with chondrite meteorites						
Element	Mantle weight %	Core weight %	Earth weight %	Condensation Temperature (K)	Carbonaceous chondrite (CI), wt%	Earth relative to chondrite
H			<0.1		1.97	
C			<0.1		3.48	
N			<0.1		0.30	
O	44.0	0	29.92		45.9	0.65
Na	0.27	0	0.18	958	0.50	0.37
Mg	22.8	0	15.50	1336	9.54	1.63
Al	2.35	0	1.60	1653	0.84	1.90
Si	21.0	6.0	16.20	1310	10.7	1.51
S	0.03	1.9	0.63	664	5.35	0.12
Ca	2.53	0	1.72	1517	0.91	1.89
Fe	6.26	85.5	31.62	1334	18.7	1.69
Ni	0.20	5.2	1.80	1353	1.09	1.65
Total	99.4	98.6	99.17			

Use the Mass Balance equation: Total mass % = (fraction in A * % of A) + (fraction in B * % of B)

Show your working in the space below:

To calculate the composition of the Earth, use the following formula:

(mantle wt% * 0.68) + (core wt% * 0.32) = Earth wt%

This is called a mass balance calculation. We weight the different components by their relative contributions to the whole. (Mantle is 68%, Core is 32% of Earth's mass.)

4) List the most abundant elements in the Earth in order of weight percent abundance:
Fe, O, Si, Mg, Ni, Ca, Al, Na, S

5) Note that in the above list, iron is more abundant than oxygen. However, in terms of the number of *atoms*, oxygen is more abundant than iron. Compare the weight percent ratio of O/Fe to the atom percent ratio of O/Fe (i.e., divide each element by its atomic mass):

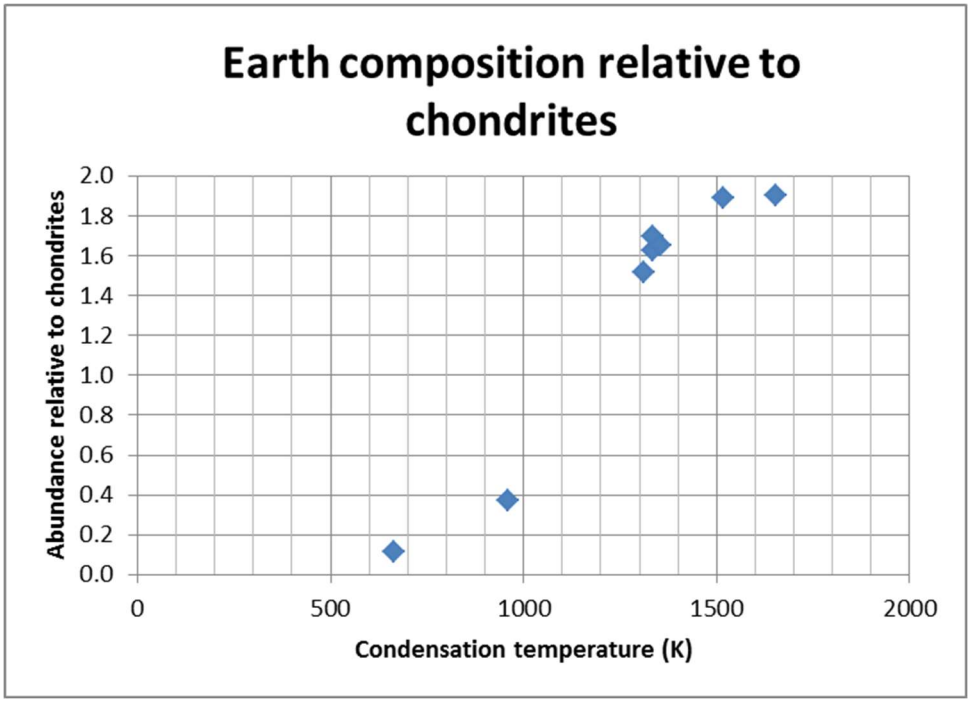
O/Fe (wt%) = 29.92/31.62 = 0.95. O/Fe (atomic) = (29.92/16) / (31.62/56) = 3.31

D. How does the composition of the whole Earth compare to the composition of the Solar System?

1) Table 4 gives the composition of carbonaceous chondrite meteorites (specifically the CI group of carbonaceous chondrites). The values are given in weight percent. These meteorites represent the average composition of the Sun and therefore the average composition of the Solar System. In the last column of Table 4, calculate the abundance of each element relative to chondrite – simply divide column 4 by column 6.

2) The column labelled "Condensation temperature" in Table 4 gives the temperature (in degrees Kelvin – K) at which the element evaporates from / condenses to a gas. Think of this as a measure of the volatility of the element. Create a graph below that shows the relationship between the condensation temperature (x axis) and the Earth composition relative to chondrite (y axis). Label the axes, including units. What is the general relationship between the two variables? *(Insert – Shapes – Circle (or whatever symbol you choose to use). Insert – Shapes – Textbox for axis labels)*

The abundance of an element in the Earth is correlated with condensation temperature. More volatile elements (lower condensation temperatures) have lower abundances in the Earth compared with chondrites. Note that on this plot, the composition of the Solar System has a value of 1 for all elements.



3) Speculate about the relationship shown on the graph. Try to form a hypothesis that would explain the relationship.

For the conditions where elements were condensing in the early Solar System, pressures were very low and condensation was mainly growth of solids directly from a gas, rather than liquids from gas.

The relationship is that with increasing condensation temperature, elements are more abundant. Stated another way, with decreasing condensation temperature, elements are less abundant. In other words, volatile elements are missing from the Earth, compared with the whole Solar System.

Describing the relationship is not a hypothesis. A hypothesis is a statement of a possible reason to explain the relationship. A hypothesis is also testable. In this case, a leading hypothesis is that when the Earth accreted (formed) it was quite close to the Sun. At this location, the disk of gas was warm enough that volatile elements had not condensed. In contrast, carbonaceous chondrites formed further from the Sun where it was cooler and more volatile elements had condensed. This hypothesis is testable – perhaps we could test it by investigating whether Mercury looks as if it formed at even higher temperatures than Earth. How else could we test it?

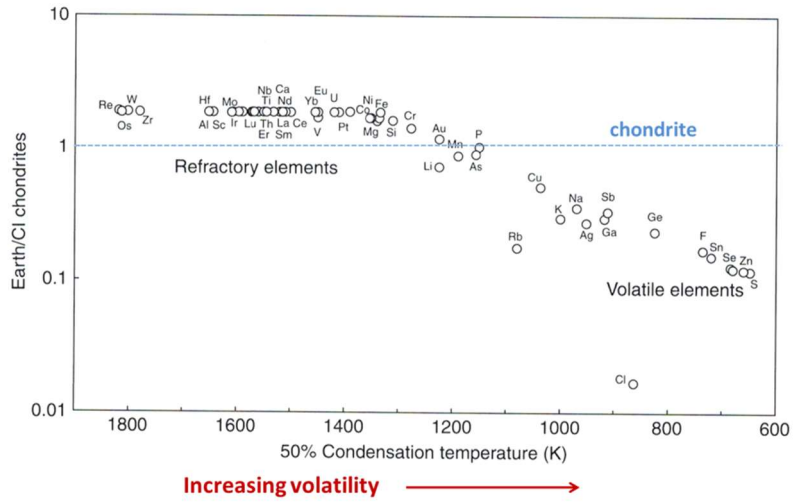
There are other reasonable hypotheses, for example perhaps the Earth had a composition similar to the Solar System when it formed, but it lost its volatile elements during the impact that formed the Moon. In a 2017 paper in Nature, Norris and Wood argue that "the pattern of volatile element depletion in the silicate Earth is consistent with partial melting and vaporization rather than with simple accretion of a volatile-rich chondrite-like body". For an extra challenge, find this paper in the library and have a go at reading it:

Norris, C. Ashley, and Bernard J. Wood. "Earth's volatile contents established by melting and vaporization." Nature 549.7673 (2017): 507-510.

Can you think of other possible hypotheses?

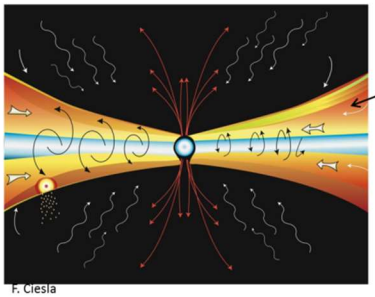
BULK COMPOSITION OF EARTH

• Overall, Earth is depleted in volatile elements compared with chondrites and the Sun.



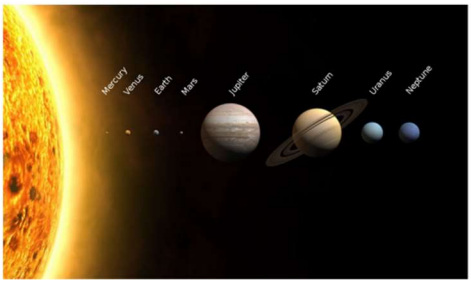
Albarede 2009 Fig. 12.11

Solar System formation



F. Ciesla

Protoplanetary disk:
Hotter closer to the Sun



For planets that formed closer to the Sun, volatile elements were still in the gas phase.